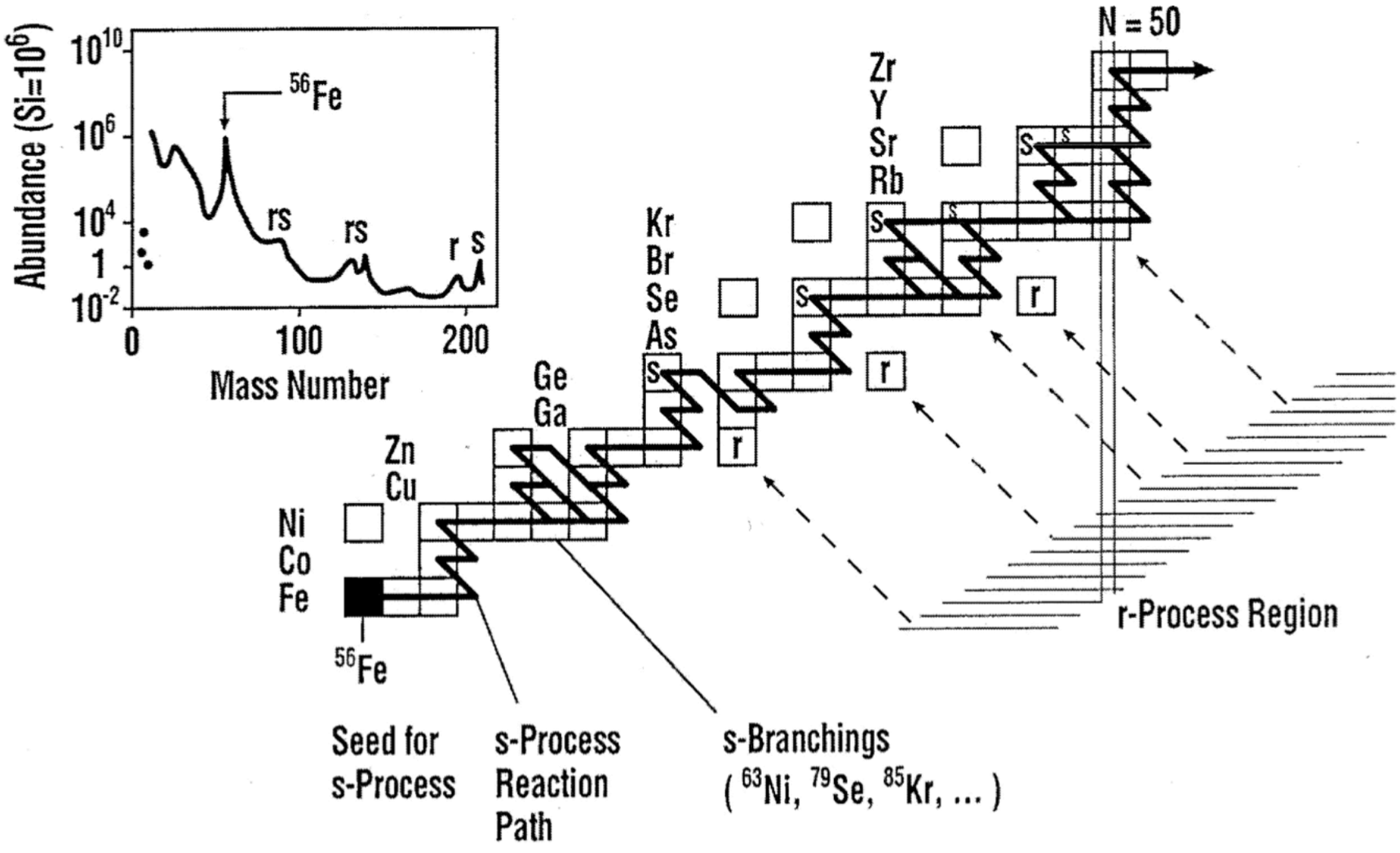

Fig. 1. Illustration of the neutron-capture processes responsible for the formation of the nuclei between iron and the actinides. Note that a *p* process has to be invoked for producing the proton-rich nuclei that are not reached by neutron-capture reactions. The schematic sketch of the solar abundance distribution in the inset indicates that the large abundance maximum in the Fe/Ni region represents the only relevant seed for the *s* process. The characteristic twin peaks in the abundance distribution between Fe and Pb result from the nuclear properties where the *s*- and *r*-reaction paths encounter magic neutron numbers.